

Java Detailed Roadmap with Subtopics & Learning Order

1. Java Basics

- History, Features, JVM/JDK/JRE
- Data Types, Variables, Operators
- Input/Output (Scanner, BufferedReader)
- Comments, Type Casting

2. Control Flow

- if/else, switch
- Loops: for, while, do-while
- break, continue, return, labeled loops

3. Object-Oriented Programming

- Classes, Objects, Constructors
- this, super keywords
- Inheritance, Polymorphism (Overloading/Overriding)
- Abstraction (abstract class, interface)
- Encapsulation, final, static
- Inner Classes, Object class methods

4. Strings & Arrays

- String vs StringBuilder vs StringBuffer
- String methods, Immutability
- 1D/2D Arrays, Arrays utility class

5. Packages & Modifiers

- Creating & Importing Packages
- Access Modifiers: public, private, protected, default

6. Exception Handling

- try, catch, finally
- throw vs throws
- Checked vs Unchecked exceptions
- Custom exceptions, Multi-catch
- try-with-resources

7. Collections Framework

- Collection Hierarchy

- List, Set, Map, Queue Implementations
- Iterator, ListIterator, Enumeration
- Comparable vs Comparator
- Collections Utility class

8. Java I/O

- File class, Byte/Character Streams
- BufferedReader/Writer
- Serialization/Deserialization
- java.nio basics

9. Multithreading

- Thread class vs Runnable
- Thread lifecycle
- Synchronization, wait/notify
- Executor framework, Deadlock handling

10. Java 8+ Features

- Lambda, Functional Interfaces
- Streams API (map, filter, reduce)
- Method References, Optional class
- Date and Time API

11. Advanced Concepts

- Generics, Reflection, Annotations
- Enums, var keyword
- Java Modules, Sealed classes
- Memory Management & Garbage Collection

12. JDBC

- JDBC Drivers, DB connection
- Statement, PreparedStatement, CallableStatement
- Transactions, Batch processing

13. Java Web Basics

- Servlets (doGet, doPost)
- JSP basics
- Spring & Hibernate Intro

14. Testing & Tools

- JUnit/TestNG, Selenium basics
- Maven/Gradle, Git, Logging frameworks